

REPORT
ON
“A Trust-Based Dynamic Routing Framework for
Enhancing Security in Software-Defined Networks”



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Abstract

Software-Defined Networks (SDN) have recently been under attack from insider threats, especially when an OpenFlow switch is compromised and behaves as a blackhole or a greyhole. This paper proposes a novel trust-based dynamic routing framework that establishes trust based on both direct behavioural attributes and indirect neighbour recommendations. The framework computes the trust rating of each switch using packet delivery ratio, delay, and drop rate, combined with neighbour recommendations. Time-based trust aging is implemented using subjective logic to formulate the Trust-Aware Constrained Shortest Path (TACSP) algorithm. Experimental results demonstrate that the framework detects a blackhole attack within 10 to 14 seconds and a greyhole attack within 30 to 60 seconds. With packet rerouting to alternative paths, it achieves a delivery ratio of nearly 100% and incurs average latency overhead of less than 10% for regular network operations.

1. Introduction

Software-Defined Networking (SDN) introduces new security challenges, as its programmable nature allows an attacker to target both the control and data planes in novel ways. Current SDN threat models focus primarily on OpenFlow switches, typically assuming that they are trustworthy. However, real-world switches may contain firmware bugs or be compromised through supply chain or insider attacks. An internal adversary who has compromised SDN switches can stealthily reside inside the network and selectively alter or block network traffic, mounting attacks within the normal boundaries of organisational trust.

SDN networks are structured around three planes: the application plane, the control plane, and the data plane. While significant work has been done to harden the control plane through authentication and encryption, data-plane security largely relies on switch authentication alone.

Network protection techniques fall into three categories: static, reactive, and trust-based. Static cryptographic mechanisms fail when endpoints are already compromised. Reactive systems such as Deep Packet Inspection and anomaly detection (including CELT-Secure, Flow-SANA, and DDoS-Auto-DMZ) identify malicious activity only after damage has occurred. Current 5G trust-based mechanisms lack support for indirect trust, real-time route reconfiguration, or data-plane integration. Reinforcement learning-based approaches fail to incorporate the historical reputation of nodes.

This paper proposes TRS, a trust-based dynamic routing framework for SDN. The framework incorporates the TACSP algorithm to proactively exclude low-trust switches from routing decisions and restore near-100% packet delivery within seconds of an attack, without human intervention.

2. Related Work

Existing SDN security research has developed defence mechanisms targeting different network layers, but these solutions largely operate independently and leave the data-plane insider attack problem insufficiently addressed.

Various frameworks assess switch trustworthiness through behavioural indicators such as packet forwarding rates, loss, delay, and subjective logic systems. However, trust values are used only for monitoring or access control at the control or application plane, not for routing enforcement. Actual flow routing continues to depend on traditional shortest-path algorithms even after a switch is identified as low-trust.

Secure routing approaches based on cryptographic methods, fuzzy logic, reinforcement learning, and optimisation operate under the assumption that all switches are trustworthy, making them vulnerable to insider attacks. Blockchain-based and machine learning-driven mechanisms applied to IoT, 5G, and AIoT networks enhance security but introduce substantial processing delays, limiting their effectiveness for real-time trust-based routing.

This work is the first to enforce trust scores directly within the path selection algorithm for SDN data-plane routing, and the first to combine trust aging, indirect recommendations, and a persistent quarantine mechanism into a unified dynamic routing solution for standard OpenFlow-based SDN environments.

Table 1: Comparison of Proposed Framework with Existing Approaches

Approach	Det. Time	Trust Routing	Indir. Trust	Rerouting
Traditional SDN	Reactive	No	No	No
CELT-Secure DPI	Post-attack	No	No	No
5G Trust Models	Moderate	Partial	No	No
Blockchain SDN	High overhead	No	No	No
TArch / RL-based	Reactive	No	No	No
Proposed (TACSP)	10-14 s	Yes	Yes	Yes

3. System and Threat Model

3.1 SDN Architecture

The proposed framework follows a standard SDN architecture with three layers. The data plane contains OpenFlow-enabled switches that forward packets based on flow table

rules. The control plane operates through a single Ryu controller that detects topology, determines routes, creates flow rules, and tracks activity via OpenFlow 1.3. The application plane provides network services through RESTful APIs.

Two topologies are used for evaluation. The linear topology (h1 to s1 to s2 to h2) tests basic attack detection. The diamond topology uses four switches (s1, s2, s3, s4), with s3 as the default path and s2 as the automatic failover when s3 is compromised.

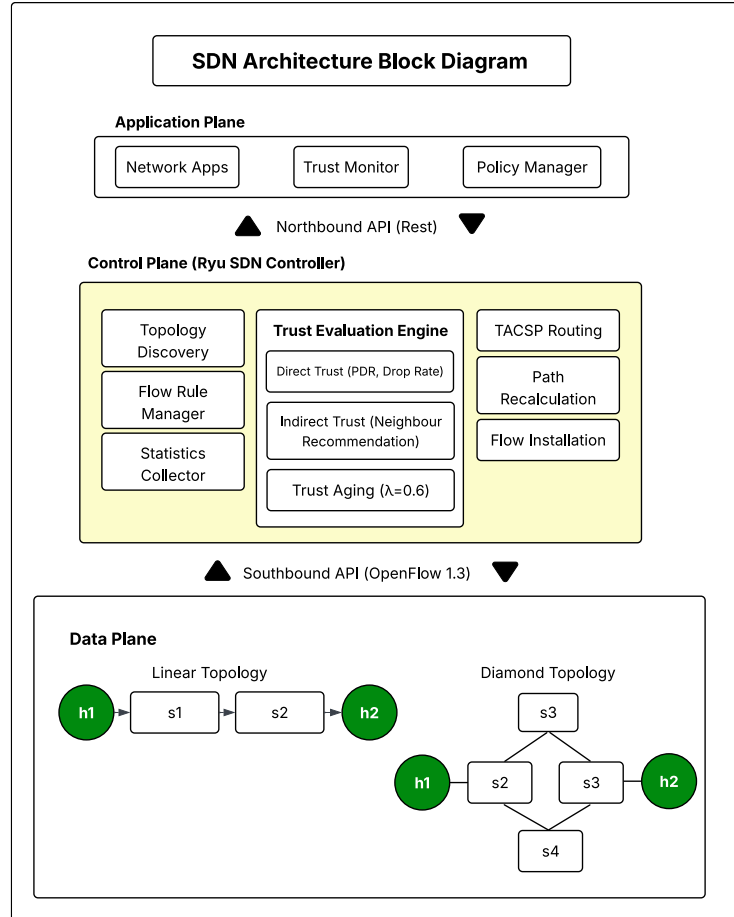


Fig. 1: Trust-Based SDN Framework Architecture

3.2 Threat Model and Attack Scenarios

The framework protects against insider attacks where an authenticated OpenFlow switch in the data plane has been compromised. The attacker manipulates packet forwarding while appearing legitimate to the controller. Two attack types are modelled:

- **Blackhole Attack:** A compromised switch installs a high-priority wildcard drop rule, causing complete packet loss. The controller detects this by observing non-zero packet counts in drop rules while forwarding entries remain inactive.
- **Greyhole Attack:** The attacker drops only a fraction of packets, maintaining partial connectivity. The controller identifies this through a sustained decline in the Packet Delivery Ratio (PDR).

3.3 Security Assumptions

The SDN controller operates in a trusted environment with secure boot and access controls. Controller-to-switch communication uses OpenFlow over TLS 1.2+ with mutual authentication. Switches with trust scores below $T_{min} = 0.5$ are removed from active routing paths.

4. Trust Evaluation Model

The framework evaluates switch trustworthiness through a hybrid trust system combining direct behavioural assessment, indirect neighbour recommendations, and a trust aging mechanism. All trust values are normalised between 0 and 1.

4.1 Direct Trust Metrics

Direct trust is computed from three controller-observed metrics: packet delivery ratio (PDR), packet delay (D), and packet drop rate (L), combined as a weighted sum:

$$T_{direct}(v) = \alpha * PDR(v) + \beta * D(v) + \gamma * L(v)$$

where $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.3$, and $\alpha + \beta + \gamma = 1.0$. In the Mininet testbed, virtual link delays are negligible, so $D(v) = 1.0$ throughout.

4.2 Indirect Trust Evaluation

Indirect trust aggregates trust reports from neighbouring switches, weighted by each neighbour's own trust value to reduce the influence of low-trust or compromised nodes:

$$T_{indirect}(v) = \frac{\sum [T(u) * T_{reported}(u,v)]}{\sum [T(u)]}$$

4.3 Trust Aging Mechanism

Trust aging applies greater weight to recent observations using exponential blending:

$$T_{aged}(v) = \lambda * T_{current}(v) + (1 - \lambda) * T_{previous}(v)$$

With $\lambda = 0.6$, the current reading contributes 60% and history 40%, smoothing transient noise while remaining responsive to sustained attacks.

4.4 Final Trust Score

The overall trust value for each switch combines direct and indirect trust:

$$T(v) = \delta * T_{direct}(v) + (1 - \delta) * T_{indirect}(v)$$

With $\delta = 0.7$, direct observation receives 70% weight. Only switches with

$T(v) \geq T_{min} = 0.5$ are eligible for routing.

5. Trust-Based Dynamic Routing Algorithm

5.1 Constrained Shortest Path Formulation

The routing problem is formulated as a Constrained Shortest Path (CSP) optimisation over a weighted directed graph $G(V, E)$. The objective is to minimise the trust-weighted path cost:

$$\text{Minimise } \sum [w(u, v) / T(v)] \text{ for each edge } (u, v) \text{ in path } P$$

Subject to three constraints: (1) all nodes in the path must satisfy $T(v) \geq T_{min}$; (2) the path must exist in the network topology; and (3) end-to-end latency must stay below L_{max} (optional). Mission-critical flows use $T_{min} = 0.75$; best-effort traffic uses $T_{min} = 0.5$. The algorithm extends Dijkstra's algorithm with trust pruning, at $O((|V| + |E|) \log |V|)$ complexity.

5.2 Trust-Aware Path Selection

The algorithm operates in two phases. Phase 1 explores the graph while excluding vertices with $T(v) < T_{min}$, pruning any vertex that fails the trust check along with all its outgoing edges. Phase 2 reconstructs the selected path by backtracking parent pointers from destination to source.

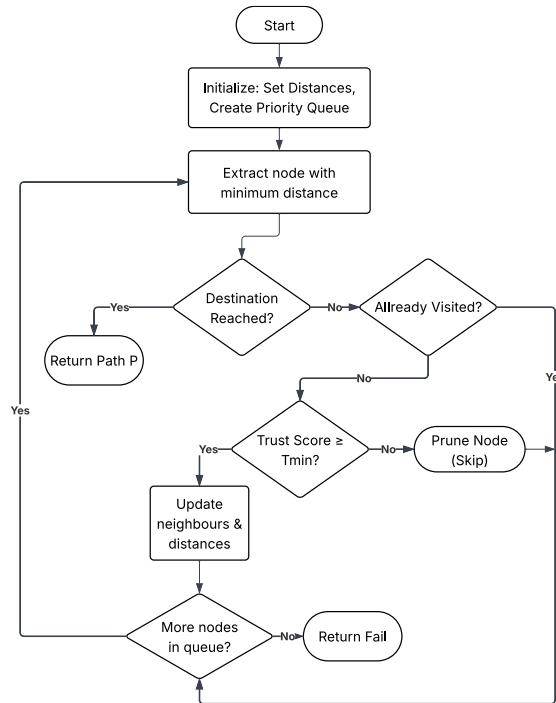


Fig. 2: Flowchart of Trust-Aware Path Selection Algorithm

5.3 Dynamic Path Recalculation

The controller triggers path recalculation under two conditions: immediately when the minimum trust on the current path falls below T_{min} (trust degradation), and periodically

to re-optimize latency when a lower-cost alternative path becomes available (trust improvement). Once the optimal path is computed, the controller installs new forwarding rules via OpenFlow FLOW_MOD messages.

6. Implementation and Experimental Setup

The framework is implemented using Mininet 2.3.0 on Ubuntu 22.04 LTS, Open vSwitch 2.17 (OpenFlow 1.3), and the Ryu SDN controller (v4.34), running on an Intel Core i3 system with 8 GB RAM. MiniNAM is used for real-time packet flow visualisation. The controller operates in reactive mode, installing flow entries on demand via PACKET_IN events.

Trust monitoring uses adaptive polling: statistics are collected every 2 seconds under normal conditions, dropping to 1 second when any switch has trust below 0.8 or an active drop rule is detected. Attack confirmation requires two consecutive suspicious polls before trust is penalised, preventing false positives from transient glitches.

Three experimental scenarios are evaluated: (1) Normal baseline operation with 100% PDR and all trust scores at 1.0; (2) Blackhole attack detection and automatic rerouting in the diamond topology; and (3) Greyhole attack detection through sustained PDR degradation.

7. Results and Performance Evaluation

7.1 Normal Operation

Under attack-free conditions, all switches maintain trust scores of 1.00 with 100% PDR and end-to-end latency of 2.0 to 2.3 ms. The trust monitoring mechanism introduces no false positives under normal forwarding conditions.

7.2 Blackhole Attack Detection and Mitigation

For the linear topology, the blackhole attack is initiated at approximately $t = 150$ s. The controller identifies the high priority drop rule within two polling intervals. The trust of the compromised switch $s1$ falls from 1.00 to 0.44 within 12 seconds and drops below $T_{min} = 0.5$ within 30 seconds, excluding $s1$ from routing at $t = 162$ s. After the attack stops at $t = 225$ s, quarantine enforcement keeps trust suppressed at 0.10 throughout the observation window.

For the diamond topology, the same attack on $s3$ triggers automatic rerouting through $s2$ via TACSP, restoring end-to-end PDR to near 1.00 within seconds. Subsequent attack cycles are handled by the same mechanism, with $s3$ permanently quarantined and $s2$ maintaining a trust score of 0.94.

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Table 2: Blackhole Attack Timeline and Recovery

Topology	Time (s)	Trust	PDR	Event
Linear	140	1.00	1.00	Normal operation
	152	0.44	0.00	Attack detected
	162	0.10	0.00	Below threshold, excluded
	225	0.10	0.68	Attack stopped; quarantine active
Diamond	140	1.00	1.00	Normal operation
	152	0.44	0.00	Attack detected
	162	0.10	0.00	Below threshold, excluded
	175	0.10	1.00	Rerouted via s2
	>438	0.10	1.00	Quarantine maintained

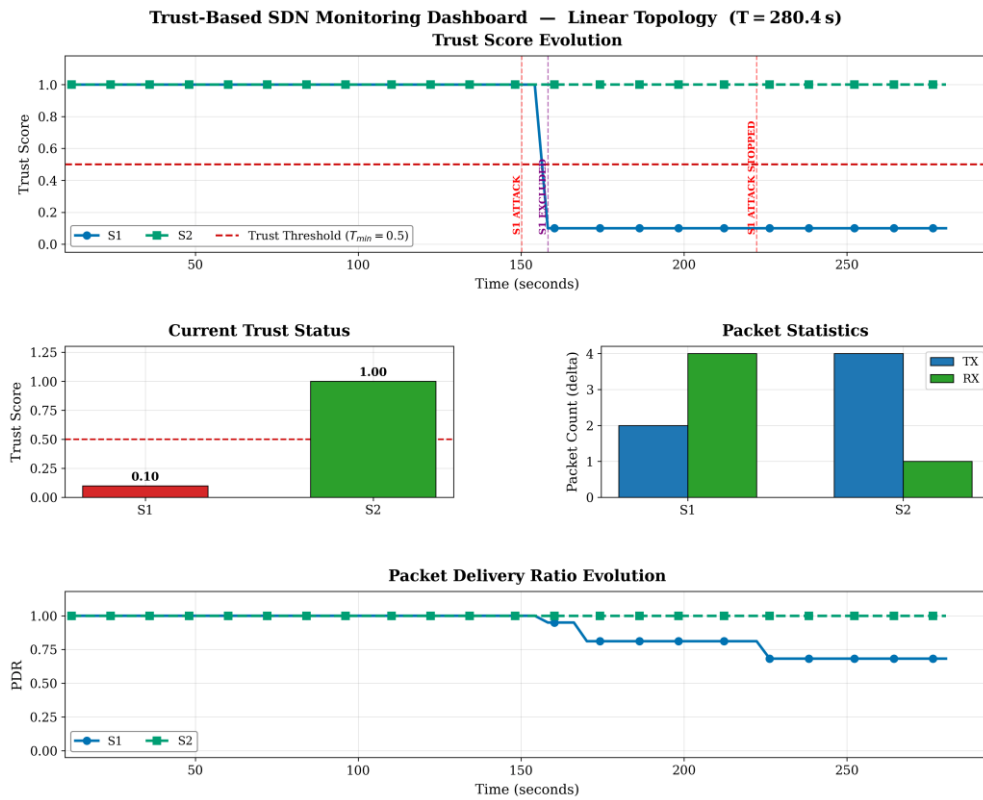


Fig. 3: Linear topology blackhole attack on s1. Attack injected at $t \approx 150$ s; s1 excluded at $t \approx 162$ s. Trust locks at 0.10 and 0.16 with some retracing and remains suppressed due to quarantine enforcement even after the attack is stopped at $t \approx 225$ s.

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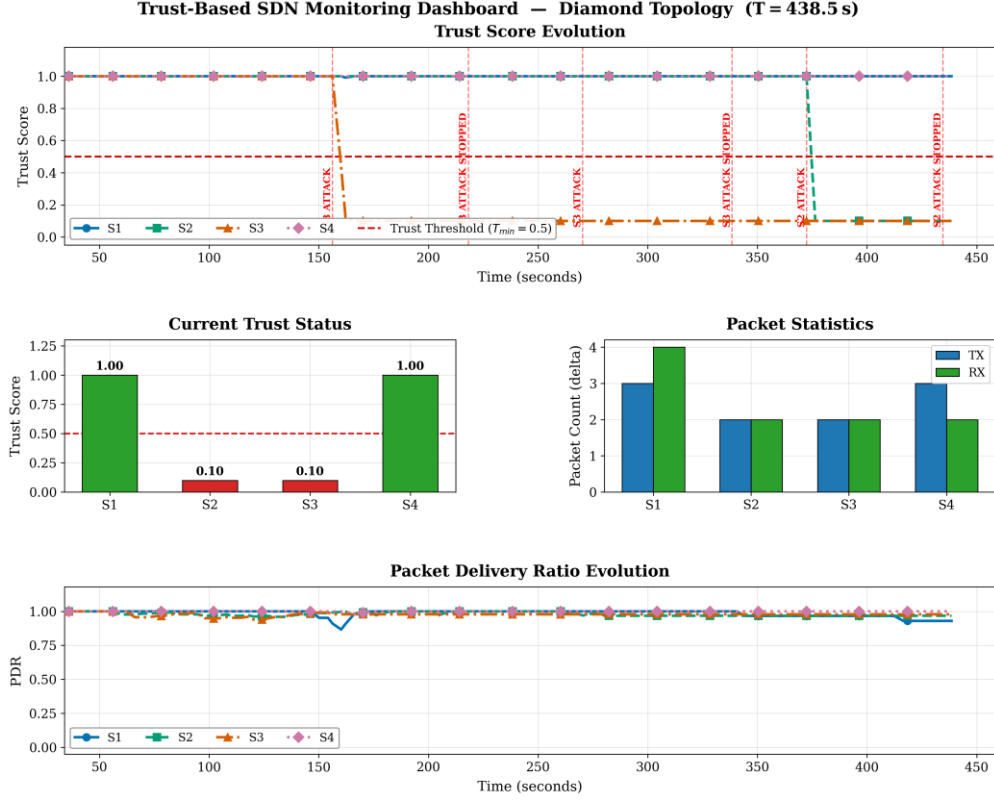


Fig. 4: Diamond topology blackhole attack demonstrating that when trust falls below a certain level, automatic rerouting to s2 occurs in order to ensure that critical functionality of the service remains unaffected. Tmin. End-to-end PDR is maintained near 1.00 throughout successive attack cycles via trust-aware path recalculation (T = 438.5 s).

7.3 Greyhole Attack Detection

Greyhole attacks produce a slow and gradual trust decay. For the linear topology under 50% packet loss, the trust of s1 declines from 1.00 to approximately 0.62 over 100 seconds, remaining above $T_{min} = 0.5$ throughout, so no rerouting is triggered. For the diamond topology under 70% loss, trust of s3 degrades to a minimum of approximately 0.60, also remaining above the exclusion threshold. End-to-end PDR on the rest of the diamond path is maintained near 1.00 throughout.

A 10-sample geometric sliding window with an 80% decay rate is used for PDR computation, producing a smooth degradation curve that identifies sustained abnormal behaviour while avoiding false positives. The trust aging factor of $\lambda = 0.6$ further prevents false threshold crossings.

Table 3: Greyhole Attack Detection Behaviour

Topology	Time (s)	Trust	PDR	Status
Linear	100	1.00	1.00	Attack start
	130	0.87	0.69	Degrading
	200	0.62	0.28	Minimum trust (above Tmin)
	360	0.75	0.29	Partial recovery; attack ongoing
Diamond	110	1.00	0.81	Attack start
	135	0.82	0.34	Degrading
	155	0.60	0.28	Minimum trust (above Tmin)
	334	0.79	0.40	Partial recovery; attack ongoing

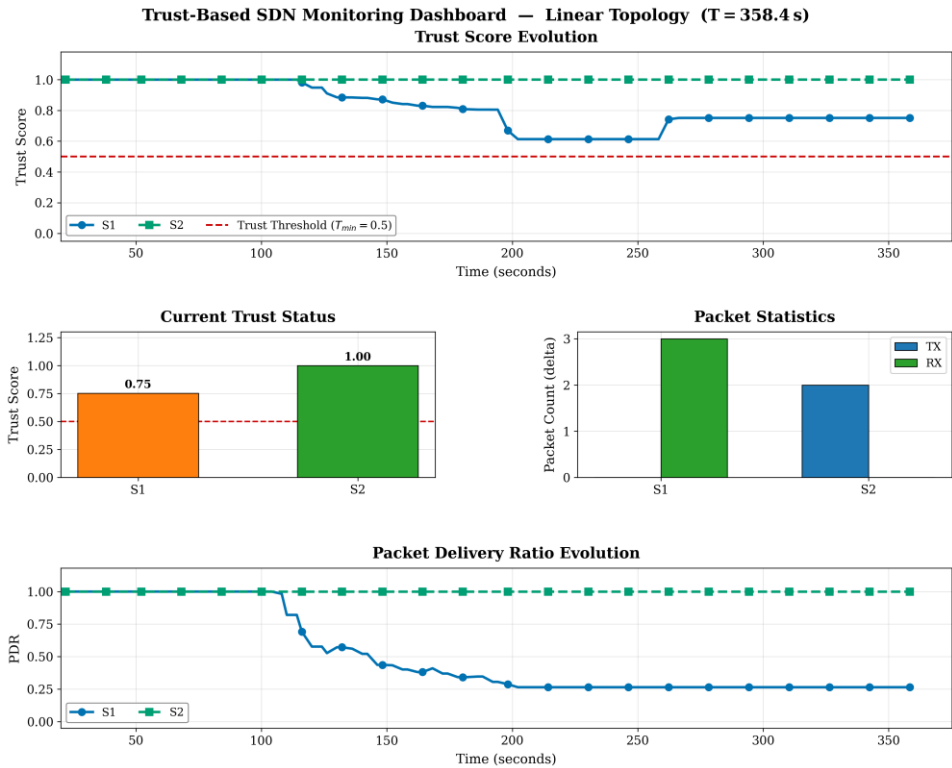


Fig. 5: Testing linear topology greyhole attack (50% loss on s1) to observe behavior under packet loss. In this scenario, over time, the trust in T (s1) continuously decreases. Starting from 1.00, it drops to approximately 0.62, remaining above $T_{min} = 0.5$ throughout. PDR stabilizes around 0.27–0.30 under sustained loss (T = 358.4s).

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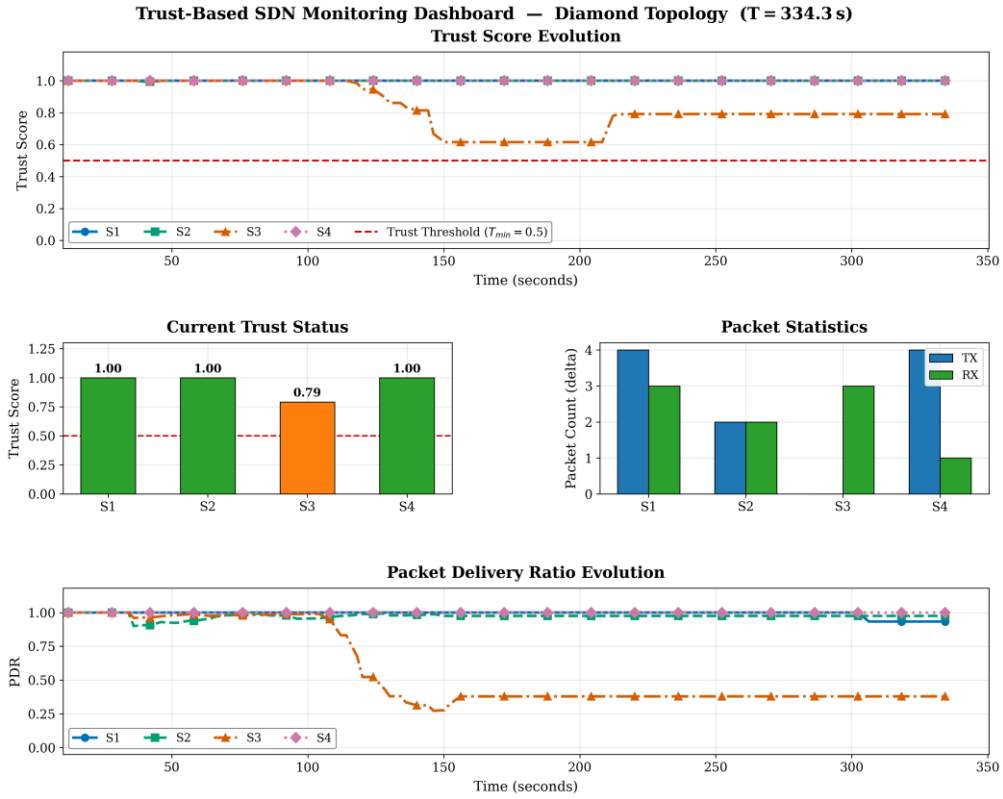


Fig. 6: Diamond topology greyhole attack (70% loss on s3) showing trust degradation on s3 to approximately 0.60, remaining above T_{min} . End-to-end PDR (s1, s2, s4) is maintained near 1.00, while s3's local PDR stabilizes at approximately 0.40 (T = 334.3 s).

7.4 Performance Summary

Table 4: Overall Performance Evaluation Results

Attack	Topology	Det. Time	Recovery	Latency
Blackhole	Linear	10-12 s	Quarantine-gated	--
Blackhole	Diamond	10-14 s	Auto reroute	+40%
Greyhole (50%)	Linear	30-60 s	Partial (above T_{min})	--
Greyhole (70%)	Diamond	30-50 s	Partial (above T_{min})	--
Normal	Linear	--	--	<10%
Normal	Diamond	--	--	<10%

8. Discussion

8.1 Proactive versus Reactive Security

Unlike traditional IDS/IPS systems that detect malicious activity only after harm has occurred, TRS prevents packet loss proactively by excluding low-trust switches from the routing search space before traffic is misdirected. In the diamond topology blackhole scenario, the controller excludes the compromised switch within 10 to 14 seconds of attack initiation and restores an end-to-end PDR of 1.00 without manual intervention. A trust recovery window allows the controller to resume forwarding through the switch once malicious activity ceases, subject to quarantine policy.

8.2 Overhead Analysis

The framework reuses existing OpenFlow flow and port statistics already collected for flow management, introducing no new southbound message types beyond occasional adaptive polling updates. TACSP path recalculation is performed only when trust falls below the threshold or at periodic optimisation intervals. Latency overhead remains below 10% during normal operation, with a higher but manageable overhead during quarantine periods.

8.3 Scalability

Trust evaluation scales with the number of switches. Direct trust computation is per-switch; indirect trust aggregation requires one pass over each switch's neighbour set at $O(|V| + |E|)$ per evaluation cycle. TACSP path computation operates at $O((|V| + |E|) \log |V|)$. The modular design, separating direct observation, indirect recommendation, trust aging, and path selection, supports extension to multi-controller deployments without structural changes to the core algorithm.

8.4 Limitations

The current implementation is validated only on Mininet-based virtual topologies with up to four switches and a single Ryu controller. Real-world performance on large-scale physical deployments with commodity OpenFlow switches and multi-controller architectures remains to be validated. Trust parameter tuning, particularly the aging factor λ and the minimum threshold T_{min} , is currently manual; automated optimisation is reserved for future work.

9. Conclusion & Future Scope

This paper presents a trust-based dynamic routing framework for enhancing security in SDN data planes against insider attacks from compromised switches. By combining direct behavioural metrics and indirect neighbour recommendations with a trust aging mechanism, and embedding these into the TACSP algorithm, the controller proactively excludes low-

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trust switches from routing decisions and dynamically recomputes paths when trust degrades. The persistent quarantine mechanism prevents compromised switches from regaining trust immediately after stopping an attack, effectively countering on-off strategies.

Experiments on Ryu and Mininet demonstrate robust detection of blackhole attacks within 10 to 14 seconds and greyhole attacks within 30 to 60 seconds, with automatic rerouting restoring near-100% packet delivery and average latency overhead below 10% during normal operation.

Future work will extend evaluation to larger multi-controller topologies and diverse traffic classes, test additional attack variants such as selective forwarding and coordinated multi-switch attacks, and explore automated trust parameter optimisation using reinforcement learning or Bayesian methods. Validation on physical hardware testbeds and integration with machine learning-based anomaly detection and blockchain-based audit logging are also identified as promising future directions.